1. C# & .NET Core

1. What are the new features of C# you have used?
2. What is Dependency Injection (DI), and how does it work in .NET Core?
3. What is the difference between AddTransient, AddScoped, and AddSingleton?
4. What is Middleware? How do you create custom middleware?
5. What is asynchronous programming? Explain async and await.
6. What is the difference between Task, Thread, and async/await?
7. What is Garbage Collection (GC) in .NET?
8. What is IDisposable, and when do you use it?
9. What is IConfiguration?
10. What is the Options Pattern?

2. ASP.NET Core Web API

11. Explain the complete request lifecycle in ASP.NET Core.
12. How do you secure a Web API?
13. How does JWT Authentication work?
14. What is the difference between Authentication and Authorization?
15. What are Filters in ASP.NET Core?
16. What is Model Validation?
17. What is API Versioning?
18. What is CORS, and why is it needed?
19. What are Action Filters and Exception Filters?
20. How do you implement Global Exception Handling?

3. Entity Framework Core

21. What is Entity Framework Core?
22. What is the difference between Code First and Database First?
23. What are Migrations?
24. What is the difference between First(), FirstOrDefault(), Single(), and SingleOrDefault()?
25. What is the difference between Eager Loading, Lazy Loading, and Explicit Loading?
26. What is LINQ?
27. How do you improve Entity Framework Core performance?
28. What is AsNoTracking(), and when should you use it?
29. How do you handle Transactions in EF Core?
30. Explain the Repository Pattern and Unit of Work Pattern.

4. SQL Server

31. What is the difference between DELETE, TRUNCATE, and DROP?

32. What is the difference between Clustered and Non-Clustered Indexes?
33. What are Stored Procedures?
34. What are Views?
35. What is a Transaction?
36. What are ACID Properties?
37. Explain different types of SQL JOINs.
38. What is a Deadlock?
39. How do you optimize slow SQL queries?
40. What is the difference between a Primary Key and a Unique Key?

5. Microservices

41. What is Microservices Architecture?
42. Why did your project move from a Monolithic architecture to Microservices?
43. How do Microservices communicate with each other?
44. What is an API Gateway?
45. How do you manage Distributed Transactions?
46. What are the major challenges in Microservices Architecture?

6. Real Project Questions

47. Explain the architecture of your Learning Management System (LMS).
48. What was the most challenging issue you solved in your project?
49. How did you improve your application's performance?
50. What was your role and responsibilities in your project?